Addressing key sustainable supply chain management issues using rough set methodology

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Abstract

Purpose – This paper aims to introduce relatively novel multi-supply chain activity overview rough set theoretic applications to aid management decision making with an especial focus on green and sustainable supply chain management.

Design/methodology/approach – The methodology is a review of recent literature with extensions around rough set or neighborhood rough set methodologies for supply chain management. An overview of how the techniques can be applied to various stages of green supply chain management, selection, evaluation, development is presented in various sections.

Findings – The paper finds that rough set methodology is flexible enough to be applied as a selection tool, performance measurement evaluation tool, and a development program evaluation tool. Its application to green supply chain management topics is warranted and valuable.

Research limitations/implications – Limitations of the approach provide additional avenues for further research. One major limitation of the research is that a real-world application to validate the approaches is necessary. Extensions and integration with other tools can also provide avenues for improvement.

Practical implications – A three-staged ecological green supplier management process may help to get a broader corporate social responsibility and general sustainability perspective on the supply chain. Management can use these tools for planning, decision making, and maintenance of green supply chain activities.

Social implications – The application of sustainability and environmental issues for supply chain management has significant social impact.

Originality/value – Methodologically, this is the first time that neighborhood rough set has been comprehensively evaluated as a tool for managing green suppliers. A comprehensive overview of the green supplier management process considering the sustainability factors helps researchers to identify many opportunities for further investigation.

Keywords Supply chain management, Performance measures, Selection, Environmental management

Paper type Research paper

Introduction

Suppliers play an important role within supply chain management. Suppliers are needed to furnish organizations with the necessary products, components, materials, and services in a timely and effective manner to help maintain a competitive advantage. Supplier decisions early in the supply chain are amplified as you move downstream, especially with the focus on green and sustainability issues. Considering green suppliers factors play a vital role for the long-term resiliency of a supply chain (Seuring and Müller,

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2008a; Ciliberti *et al.*, 2008; Zhu *et al.*, 2008). Given this rise in managerial and sustainability importance, strategic management of green suppliers becomes important and commodity and price based supplier relationships are no longer acceptable for green suppliers and organizations that seek to introduce innovative long-term supplier management issues.

Managing suppliers can take a general three staged process, selecting, evaluating, and then further developing these suppliers. In this paper we introduce each of these major steps for green supplier management. Within this three stage context we introduce a relatively novel rough set theoretic application to aid in this process, with a special focus on green and sustainable supply chain management. The work here can be found in more detail in Bai and Sarkis (2010a, b) and Bai *et al.* (2010). The overview review provides a logical development of these tools and their interrelationships and the flexibility of the rough set methodology as applied to green supply chain management issues.

The paper begins with an overview of the various concerns for each of these stages. Rough set theory notation and methodology is then introduced. Exemplary rough-set-based techniques that can be utilized to help management decision makers at each stage are detailed in the next section. At last conclude this paper with future directions for additional research.

Managing green suppliers

Supplier selection

Researchers and practitioners have focused on green supplier management and development because they realize the importance of greening suppliers in a world where environmental concerns are also competitive concerns. One research stream focuses on supplier selection and supplier evaluation. The existence of models and tools for environmental consideration in supplier selection research is still emerging, while more general sustainability issues, incorporating other social sustainability dimensions into supplier selection are quite scarce (Hutchins and Sutherland, 2008).

Increasingly, researchers are addressing supplier selection issues integrating environmental aspects (Handfield *et al.*, 2002; Humphreys *et al.*, 2003; Lee *et al.*, 2009; Sarkis, 2006). Broader sustainability (triple-bottom-line) factors may also be integrated into the selection factors and include economic/business, environmental and social dimensions. Thus, the breadth and diversity of selection factors make the supplier selection process even more cumbersome and complex. Using historical performance of suppliers and previous decisions related to these suppliers are critical information that many approaches have not effectively integrated into their decision models.

Environmental and economic supplier performance measurement

Some literature has established the organizational need to consider developing performance measurement systems (PMS) for their supply chains (Beamon, 1999). Environmental performance measurement for the supply chain was concurrent to the early stages of supply chain PMS development (McIntyre *et al.*, 1998). Given that the importance of sustainability criteria to measure and report supply chain performance has increased in today's competitive and social environment (Vasileiou and Morris, 2006). Effective performance measurement systems and metrics development is also critical for making the business case for sustainable supply chains (Keating *et al.*, 2008).

Practical difficulties with general PMS and supply chain PMS include the need to effectively manage the dynamic nature of performance measures and metrics used within these systems (Wickramatillake *et al.*, 2007). The integration of tangible and intangible performance measures add to the complexity of managing supply chain performance

(Giannakis, 2007). Industry is always looking to see how they can become more efficient. Having cumbersome and complex supply chain performance measures diminishes the management efficiency of the supply chain. Managing key performance indicators (KPI) is critical for supply chain managers (Chae, 2009). Formal modeling based on analytical tools for supply chain KPI determination have been lacking. An important issue is integrating environmental performance measures into these evaluations. Another issue is the determination and management of these performance measures. Especially important is how to determine which performance measures are effectively monitoring the expected outcomes of management. Monitoring and managing ecological and business/economic sustainability of supply chains adds greater complexity to PMS management. Therefore, within supply chain management supplier performance measures management is one of the critical issues faced by operations and purchasing managers to help organizations maintain a strategically competitive position.

Green supplier development

The third stage in our process for managing green supply chains is managing suppliers' continuous improvement efforts (development). Supplier development approaches and practices have been extensively proposed and documented in the literature. These studies investigate various supplier management development practices. There is significant opportunity for formal models and decision support tools that organizations can utilize to help them improve and systematize their supplier development programs (Simatupang and Sridharan, 2004; Gunasekaren *et al.*, 2000). The extant literature has not considered any prioritization to address the "*locus* of investments" for supplier development initiatives (Narasimhan *et al.*, 2008).

There is a research gap on how an organization can effectively manage supplier development programs, and this gap is larger for green supplier development. The use of formal models to aid green supplier development management is virtually non-existent. A critical aspect of this inter-organizational focus is the development of suppliers, many who are small and do not have the necessary resources to address serious environmental issues they face (Seuring and Müller, 2008b). Even though significant research has been initiated and completed on green supply chain management (Seuring and Müller, 2008c; Srivastava, 2007; Ilgin and Gupta, 2010), the lack of general supplier development literature and research related to business performance is under-researched. Much of this research is empirically based with a focus on general relationships between supplier development practices and performance (e.g. Narasimhan *et al.*, 2008; Krause *et al.*, 1998; Wagner and Krause, 2008).

Even though other supply chain management research issues may be addressed, these three stages, selection, performance evaluation, supplier development, are important aspects to green supplier management. We do not address managerial issues such as managing the risk of these suppliers, dropping suppliers, and involvement of suppliers in internal operations, for example. But, these are all additional potential areas for evaluation using rough sets. At this time, we show how rough set theory can be applied to management at the three stages we identify in this paper.

Basic rough set theory

Basic rough set theory classifies objects into similarity classes (clusters) containing objects that are indiscernible with respect to previous knowledge. Similarity classes are employed to determine hidden patterns within the data. Applications of basic rough set theory have primarily occurred in data mining approaches. Integrated with rough set theory is approximation vagueness and is defined by precise values of lower and upper approximations. Lower approximations describe the domain objects which definitely belong to the subset of interest. Upper approximations describe objects which may possibly belong to the subset of interest. The difference between the upper and the lower approximations constitutes a boundary region for the vague set. Hence, basic rough set theory expresses vagueness by employing a boundary region of a set. If the boundary region of a set is empty it means that the set is crisp, otherwise the set is rough (inexact).

Some definitions and notation for basic rough set follows:

Definition 1

Let U be the universe and let R be an equivalence relation on U. For any subset $X \in U$, the pair $T = (U, R)$ is called an approximation space.

The two subsets:

$$\underline{R}X = \{x \in U | [x]_R \subseteq X\} \tag{1}$$

$$\bar{R}X = \{x \in U | [x]_R \cap X \neq \phi\} \tag{2}$$

are called the R-lower (1) and R-upper (2) approximation of X, respectively.

$$BN_R(X) = \bar{R}X - \underline{R}X \tag{3}$$

Thus the R-boundary region of X is represented by expression (3). If $BN_R(X) = 0$, then we have a crisp set, a $BN_R(X) > 0$ provides us with a rough set for our evaluation. $POS_R(X) = \underline{R}X$ is used to denote R-positive region of X (represented by the blackened cells in Figure 1). $NEG_R(X) = U - \bar{R}X$ is used to denote R-negative region of X (which is represented by the white cells in Figure 1). The cells in Figure 1 represent objects to be evaluated, white cells are considered to be outside the rough set, black cells are definitely within the rough set. Gray cells in Figure 1 may or may not fit within the set.

Green supplier selection may use a triple-bottom-line approach. Given the complexity and volume of sustainability factors and attributes, identifying which attributes to utilize and their relative influences on current and previous decisions will be a great aid to managers in their decision making. The data-mining characteristics of rough set theory help to advance this development along with the capability of effectively considering the tangible and intangible factors. We now introduce our rough set based selection model with these sustainability attributes to select the preferable green suppliers.

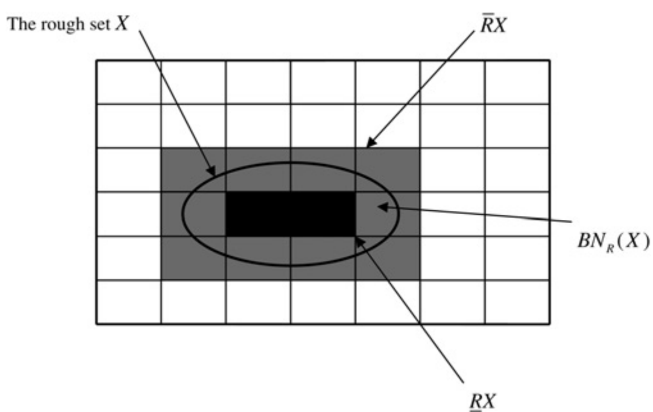

Figure 1.
A graphical representation of a rough set environment

Sustainability supplier selection

Setting up a basic rough set table, the selection model proposed is composed.

First, let us assume that a database of suppliers exists. This table is defined by $T = (U, A, V, f^{\otimes})$, where $U = \{S_1, S_2, \ldots, S_m\}$ is a set of m alternative suppliers called the universe. $A = \{a_1, a_2, \ldots, a_n\}$ is a set of n attributes for the suppliers. Where the $f^{\otimes}$ is a function used to define the values V. $T = (U, A, V, D, f^{\otimes})$ and D is a specific set of historical attributes representing previous decisions. For the illustrative case $U = \{S_i, i = 1, 2, \ldots, 30\}$ (i.e. 30 suppliers) with nine attributes $A = \{a_j, j = 1, 2, 3, \ldots, 9\}$ each. We assume those nine attributes three environmental attributes, Ev1, Ev2, and Ev3; three economic/business attributes Ec1, Ec2, and Ec3; and three social attributes So1, So2, and So3 also. An example decision table is shown in Table I.

The decision maker evaluates the 30 suppliers, on each of the nine sustainability attributes. Notice that the final two columns of Table I, data on previous historical decisions (D vector) on whether a supplier was previously selected.

The decision values are categorized into three levels $\{$*yes, yes and no, no*$\}$. These two variables represent previous histories and the resulting decision, where *yes* would represent a history of always selecting this supplier whenever they were considered, *yes and no* would be suppliers who were sometimes selected and sometimes not selected historically, and a *no* decision would represent suppliers who were never selected when they were considered. More delineations could be provided, but these three cover the spectrum of decisions.

We shall assign values of $\{2, 1, 0\}$ to the $\{$*yes, yes or no, no*$\}$ categories, respectively. Using the general rough set expression (1) let the lower approximation of preferred suppliers for a specific decision variable x (D_i^x) (S^*) be defined as:

$$\underline{R}S_* = \{S_i \in U | [S_i]_R \subseteq S_*\} \tag{4}$$

where $S_*^x = \{S_i^x | D_i^x = yes\}$.

Let us begin with the situation where we are considering the D_i^2 as the historical selection decisions from Table I (which incorporates all sustainability factors). Using

Supplier	Ev1	Ev2	Ev3	Ec1	Ec2	Ec3	So1	So2	So3	Previous decisions D
	Decision maker									
1	G	SG	SF	SG	SG	P	SF	SF	SF	1
2	SF	VP	G	VP	SF	VG	SG	SG	VP	0
3	G	VG	SF	G	G	G	SF	SG	P	2
4	VG	G	P	G	SG	G	G	F	G	2
5	G	VG	G	P	G	G	P	P	P	2
6	VG	SG	VP	G	SG	SG	VG	G	G	2
7	SG	F	F	VP	VP	G	G	P	VP	0
8	P	G	F	P	P	G	VG	G	VP	1
9	P	VP	VP	VP	P	G	P	G	G	0
10	G	P	VP	VP	G	G	P	P	G	0
11	VP	G	G	SF	G	VP	VP	P	G	0
12	G	G	G	G	SG	G	G	G	VG	2
**	...	...	...	...	...	...	...	...	...	...
28	G	F	SF	P	G	P	G	SG	SG	1
29	SG	P	G	P	F	P	VP	P	SF	0
30	SF	G	P	SF	P	SG	SG	P	F	0

Table I. Supplier evaluations on sustainability factors by decision makers and historical selection decisions

Table I, we arrive at the partitioned sets of the suppliers. That is $[S_i]_R = \{S_1,S_{17},S_{20}\}$, $\{S_2\},\{S_3\}, \{S_4\}\{S_5\}, \{S_6\},\{S_7\},\{S_8\},\{S_9\},\{S_{10}\},\{S_{11}\},\{S_{12}\},\{S_{13}\}, \{S_{14}\},\{S_{15}\}, \{S_{16}\}$, $\{S_{18},S_{23},S_{28}\},\{S_{19}\},\{S_{21}\},\{S_{22}\},\{S_{24}\},\{S_{25}\},\{S_{26}\},\{S_{27}\},\{S_{29}\}, \{S_{30}\}\}$, which represent unique attribute values for every subset. For example $\{S_{18}, S_{23}, S_{28}\}$ represent three suppliers that have the same values for each of the sustainability attributes. We can use these partition sets to help complete our analysis for determining the set of preferable suppliers.

The initial subset $S_* = S_*^2 = \{S_i^2|D_i^2 = yes\}$ is formed and will be used to determine the final set of preferable sustainable suppliers based on the sustainability attributes and the previous decisions. $S^* = \{\{S_3\}, \{S_4\}, \{S_5\}, \{S_6\}, \{S_{12}\}, \{S_{17}\}, \{S_{18}\}, \{S_{20}\}, \{S_{23}\}, \{S_{27}\}\}$ is obtained.

We now remove those suppliers whose attribution values are the same as those whose decisions $D_i^2 = 0$ or 1. This removal results in the lower approximation set ($\underline{R}S^* = \{S_i \in U|[S_i]_R \subseteq S^*\}$). This lower approximation set leaves no vagueness as the preference of these suppliers based on historical decisions. We then obtain $\underline{R}S^* = \{\{S_3\},\{S_4\},\{S_5\},\{S_6\},\{S_{12}\},\{S_{27}\}\}$. Therefore suppliers S_3, S_4, S_5, S_6, S_{12}, and S_{27} can be viewed as the final set of suppliers that could be considered preferable when evaluating all the sustainability attributes and the previous historical decisions.

To determine the most preferred suppliers a gray-relation-TOPSIS based approach (Bai and Sarkis, 2010a) that assigns values to these supplier evaluations and then compares the difference from the ideal. We do not go further at this stage, but only show how rough set can help identify the narrowed set of preferable suppliers for supplier selection.

The results are shown in Table II. We see that supplier 12 using a modified gray-relation-TOPSIS approach with rough set theory is the preferred supplier.

In this situation, rough set theory allows for distillation of a larger set of suppliers into a smaller set of candidate preferred suppliers, and eventually the selection of preferred supplier. Rough set theory application to supplier selection and decision making contributes through use of historical decisions integrating previous organizational knowledge and learning into the latest decision process. As time progresses, organizations can further refine their decision making quality to either maintain some consistency and/or improve their decision process with further weighting and development of attributes that are salient for the organization's strategic direction. This approach helps in this process.

Decision attribute set (**A**)	Decision variable (*D*)	Initial partitioned set $[S_i]_R$	R-upper approximation ($\bar{R}X$)	Rough set (*X*)	R-lower approximation ($\underline{R}X$)	Preferred supplier
EV1,EV2, EV3, EC1,EC2, EC3,SO1, SO2,SO3	*D*	$\{S_1,S_{17},S_{20}\}$, $\{S_2\}$, $\{S_3\}$, $\{S_4\}$ $\{S_5\}$, $\{S_6\}$, $\{S_7\}$, $\{S_8\}$, $\{S_9\}$, $\{S_{10}\}$, $\{S_{11}\}$, $\{S_{12}\}$, $\{S_{13}\}$, $\{S_{14}\}$, $\{S_{15}\}$, $\{S_{16}\}$, $\{S_{18}, S_{23}, S_{28}\}$, $\{S_{19}\}$, $\{S_{21}\}$, $\{S_{22}\}$, $\{S_{24}\}$, $\{S_{25}\}$, $\{S_{26}\}$, $\{S_{27}\}$, $\{S_{29}\}$, $\{S_{30}\}$	S_1, S_3, S_4, S_5, S_6, S_{12}, S_{17}, S_{18}, S_{20}, S_{23}, S_{27}, S_{28}	S_3, S_4, S_5, S_6, S_{12}, S_{17}, S_{18}, S_{20}, S_{23},S_{27}	S_3, S_4, S_5,S_6, S_{12}, S_{27}	S_{12}

Table II.
Summary results rough set methodology for sustainable supplier selection using an adjusted TOPSIS approach

Rough set theory for green supplier development

In this portion of the paper we use rough set theory and methodology with incomplete information.

The illustrative case example will provide us with insights into application of the rough set approaches for green supplier development program evaluation. First, let us define the information system table for the illustrative application. For our illustrative case $U = \{S_i, i = 1, 2, \ldots, 30\}$ (i.e. 30 suppliers) with ten total conditional attributes $C = \{c_j, j = 1, 2, 3, \ldots, 10\}$ each. The conditional attributes represent various supplier development programs and organizational characteristics. We assume the buyer organization has a mixture of three green knowledge transfer conditional attributes, G1, G2, and G3; two investment and resource transfer conditional attributes I1, I2; and two management and organizational practices attributes M1, M2. We also incorporate three organizational characteristic attributes such as profitability, size, and industry (O1, O2, and O3, respectively). These environmental supplier development attributes are a categorization of environmentally oriented (green) supplier development practices and activities.

For the case illustration we assume two decision attributes $D1$ and $D2$. $D1$ represents a column whose outcome (decision attribute) is environmentally oriented performance. $D2$ represents a column whose outcomes are business-oriented performance. Table III shows the full information system table (values under the conditional attributes range from 1 (not involved) to 5 (very highly involved); values in the decision attributes represent 1 (low performance) to 3 (high performance).

One final aspect of our information system table is that not all information for each attribute across the various suppliers may be available. Missing, or incomplete, information is designated by an "*" in Table III. We incorporate incomplete data to show the flexibility of the approach and to introduce some of the latest methodological characteristics of rough set theory.

Next we identify the information significance of each of the attributes (green supplier development programs or organizational characteristics). The information significance will help to reduce the set of programs and organizational characteristic attributes and only keep those conditional attributes that provide strong evidence and relationships with the performance outcomes. This relationship may be negative or positive. We now form an initial matrix made up of only the "core" remaining conditional attributes and the decision attributes. In this situation $core(C) = \{G1, M1, M2, O2, O3\}$.

	Conditional attributes (C)										Decision attributes	
Supplier identifier	G1	G2	G3	I1	I2	M1	M2	O1	O2	O3	D1	D2
Supplier 1	5	*	5	3	4	5	5	3	3	2	3	3
Supplier 2	2	3	1	4	1	*	4	3	2	1	1	2
Supplier 3	5	5	3	2	3	4	4	3	3	3	3	3
Supplier 4	2	3	1	4	1	2	2	3	2	1	1	2
Supplier 5	4	5	2	2	5	3	4	2	3	2	2	2
Supplier 6	2	5	3	2	4	5	5	1	2	2	3	1
Supplier 7	2	5	3	2	4	5	5	1	2	2	3	1
Supplier 8	5	2	3	4	3	*	4	1	2	3	*	2
Supplier**	...	...	...	...	...	...	...	...	...	...	...	...
Supplier 28	4	*	1	3	2	1	2	2	3	3	1	3
Supplier 29	3	2	5	4	3	2	3	1	2	1	2	1
Supplier 30	1	1	2	5	2	1	1	3	1	2	1	2

Table III. The incomplete information system for the illustrative application

We take the core attributes (initial reducts) of the previous steps and relate them to the various decision attributes through a series of rules to discern the relationships and further reducts. We will complete this step by utilizing the rough set exploration system (RSES) (Bazan and Szczuka, 2005) software and uses expressions below based on works by Kryszkiewicz (1998, 1999), Liang *et al.* (2006), and Pawlak (1991).

The full set of discernibility rules for suppliers with high environmental, $\Delta^{D1=3}$ (i.e. $D1 = 3$), and high business, $\Delta^{D2=3}$ (i.e. $D2 = 3$), performance are shown in Table IV. The first rule, $(G1 = 5) \wedge (O2 = 3)$, states that if a supplier has a high level of involvement in green knowledge transfer program 1 ($G1 = 5$) and has high profitability ($O2 = 3$) then the environmental performance is expected to be high. This rule is relatively strong with the RSES program generating three observations of this rule.

Initially, to arrive at a slightly more concentrated set of relationship rules, we go through a simple "minimum decision algorithm." If there are relationship rules (functions) that have the same set of conditional attributes within these relationships, then it would suffice the rule to take the lower bound, encompassing values associated with those attribute relationships. This is a dominance rule that assumes ordinal variables.

Another consideration in reducing the set of rules is to determine "infeasible" rules. Even the initial rules generated may miss some lower level performance results that cause infeasibility to occur. To apply this rule reduction technique, only ordinal attributes G1; M1; M2; O2 may be used. Attribute O3 is ordinal thus, the "bigger is better" dominance characteristic does not work on this rule reduction technique. The final rule set for the $D1$ and $D2$ decision results are shown in Table V.

But, some organizations will also be concerned with arriving at a "win-win" situation that provides rules for both high environmental and high business performance. To arrive at these rules one more step in this evaluation process is completed to integrate the final rule sets. The final rule set for the generalized performance (joint economic and environmental performance) is shown in Table VI.

For this organizational illustrative application (final results in Tables V and VI), the major missing conditional attributes are green supplier development programs from the investment and resource transfer categories. The organization should evaluate why these programs are not producing either good environmental or business performance.

Table IV.
Initial full set of rough set decision rules (Δ^{D}) generated for $D = D1$, $D2$, ($D1 = 3$, $D2 = 3$) (high performance)

$\Delta^{D1=3}$	Number of occurrences	$\Delta^{D2=3}$	Number of occurrences
$(G1 = 5) \wedge (O2 = 3)$	3	$(O2 = 3) \wedge (O3 = 3)$	5
$(M2 = 4) \wedge (O2 = 3) \wedge (O3 = 3)$	3	$(G1 = 5) \wedge (O2 = 3)$	3
$(M1 = 4)$	3	$(G1 = 4) \wedge (O3 = 3)$	3
$(M1 = 5) \wedge (O2 = 3)$	2	$(G1 = 4) \wedge (M1 = 1)$	2
$(G1 = 2) \wedge (M1 = 5)$	2	$(M1 = 1) \wedge (M2 = 2)$	2
$(G1 = 2) \wedge (M2 = 5)$	2	$(M1 = 1) \wedge (O2 = 3)$	2
$(G1 = 5) \wedge (M1 = 5)$	1	$(M1 = 1) \wedge (O3 = 3)$	2
$(G1 = 5) \wedge (M2 = 5)$	1	$(G1 = 4) \wedge (M2 = 2)$	2
$(G1 = 5) \wedge (O3 = 2)$	1	$(M2 = 2) \wedge (O2 = 3)$	2
$(M2 = 5) \wedge (O2 = 3)$	1	$(M2 = 2) \wedge (O3 = 3)$	2
$(G1 = 4) \wedge (M1 = 5)$	1	$(G1 = 5) \wedge (M1 = 5)$	1
$(M1 = 5) \wedge (M2 = 4)$	1	$(G1 = 5) \wedge (M2 = 5)$	1
$(M1 = 3) \wedge (O3 = 3)$	1	$(G1 = 5) \wedge (O3 = 2)$	1
$(G1 = 4) \wedge (M2 = 4) \wedge (O3 = 3)$	1	$(M2 = 5) \wedge (O2 = 3)$	1
		$(M1 = 3) \wedge (O3 = 3)$	1

Practically, they can either seek to improve these green supplier development programs or eliminate them.

Using rough set analysis, general rules are developed to help guide either practicing managers or researchers for more in depth evaluation of these initial, rough, results. In summary, managers and researchers can use this technique to at least:

- determine which programs to emphasize;
- determine which programs to de-emphasize;
- what characteristics contribute to performance; and
- where they could put additional investment or disinvestment in their suppliers.

Green performance measurement of suppliers neighborhood rough set theory

We now describe how green performance measurement using neighborhood rough set can be completed. First we introduce the concept of neighborhood rough sets.

Neighborhood rough sets

For neighborhood rough sets, the determination of what should be included in a rough set is allowed to have some flexibility based on a distance relationship for the attributes. The definitions and methodology presented in this paper for neighborhood rough sets are based on the developments of Hu *et al.* (2008a, b) and incorporate grey based measures.

Definition 2. Given an arbitrary $x_i \in U$ and $B \subseteq C$, the neighborhood $\delta_B(x_i)$ of x_i in attribute space B is defined as:

$$\delta_B(x_i) = \{x_j | x_j \in U, \Delta_B(x_i, x_j) \leq \delta\}, \tag{5}$$

where Δ is a distance function.

The rules for $D1 = 3$	Number of occurrences	The rules for $D2 = 3$	Number of occurrences
(G1 = 2) ∧ (M1 = 5)	2	(G1 = 5) ∧ (O2 = 3)	3
(G1 = 2) ∧ (M2 = 5)	2	(G1 = 5) ∧ (M1 = 5)	1
(G1 = 5) ∧ (O2 = 3)	3	(G1 = 5) ∧ (M2 = 5)	1
(G1 = 5) ∧ (O3 = 2)	1	(G1 = 5) ∧ (O3 = 2)	1
(G1 = 4) ∧ (M2 = 4) ∧ (O3 = 3)	1	(M2 = 5) ∧ (O2 = 3)	1
(M1 = 3) ∧ (O3 = 3)	1	(O2 = 3) ∧ (O3 = 3)	5
(M1 = 5) ∧ (O2 = 3)	2		
(M2 = 4) ∧ (O2 = 3) ∧ (O3 = 3)	3		
(M2 = 5) ∧ (O2 = 3)	1		

Table V. Final set of rough set decision rules (Δ^{D}) generated for $D = D1$, $D2$, ($D1 = 3$, $D2 = 3$) (high performance)

(G1 = 5) ∧ (O2 = 3)
(G1 = 5) ∧ (O3 = 2)
(M2 = 5) ∧ (O2 = 3)
(G1 = 5) ∧ (M1 = 5)
(G1 = 5) ∧ (M2 = 5)
(M2 = 4) ∧ (O2 = 3) ∧ (O3 = 3)

Table VI. Final set of rough set decision rules ($\Delta^{G=3}$) generated for joint high performance environmental and business outcomes

Definition 3. Let $\otimes v_{ij}$ denote the gray based value (fuzzy value; neighborhood rough sets do not need to have gray numbers and can use regular scalar numbers) of object i on attribute j, then using:

$$\Delta_p(x_i, x_k) = \left(\sum_{j=1}^{N} |[(\underline{v_{ij}} - \underline{v_{kj}})^2 + (\overline{v_{ij}} - \overline{v_{kj}})^2]^{1/2}|^p \right)^{1/P} \tag{6}$$

when $p = 1$ it is defined as a Manhattan distance; $p = 2$ is a Euclidean distance; and $p = \infty$ is a Chebychev distance (Wilson and Martinez, 1997). The Chebychev distance equation for gray scale is represented by:

$$\Delta_p(x_i, x_k) = \max_j |(\underline{v_{ij}} - \underline{v_{kj}})^2 + (\overline{v_{ij}} - \overline{v_{kj}})^2]^{1/2}| \tag{7}$$

Definition 4. Given a neighborhood decision system NDS = <*U*,*C* ∪ *D*,*N* >, where N is a neighborhood relation on U, $X_1, X_2, \ldots, X_N$ are the object subsets with decisions 1 to N, and $\delta_B(x_i)$ is the neighborhood information granule generated by attributes $B \subseteq C$, the lower and upper approximations of decision D with respect to attributes B are defined as:

$$\underline{N_B}D = \cup_{i=1}^{N} \underline{N_B}X_i, \overline{N_B}D = \cup_{i=1}^{N} \overline{N_B}X_i \tag{8}$$

where:

$$\underline{N_B}D = \{x_i | \delta_B(x_i) \subseteq X, x_i \in U\},\ \overline{N_B}D = \{x_i | \delta_B(x_i) \cap X \neq \emptyset, x_i \in U\}. \tag{9}$$

The decision boundary region of D with respect to attributes B is defined as:

$$BN(D) = \overline{N_B}D - \underline{N_B}D \tag{10}$$

There are two key factors that define a neighborhood: the distance calculation (p value) and the threshold value δ. When $\delta = 0$ the objects in the neighborhood granule are equivalent and the neighborhood rough set model reduces to the basic rough set model. Therefore, neighborhood rough sets (expressions (8)-(10)) are generalized forms of classical rough sets (expressions (1)-(3)).

Definition 5. Neighborhood rough sets can also be described and generalized by introducing a measure of inclusion (Hu *et al.*, 2008a; Ziarko, 1993). Given two sets A and B in the universe U, we define A's inclusion degree in B as:

$$I(A, B) = \frac{|A \cap B|}{|A|}, \quad \text{where} \quad A \neq \emptyset. \tag{11}$$

where $|*|$ is the cardinality of a set. Incorporating expression (11) into the identification of a neighborhood rough set, we now define the following.

Definition 6. Given any subset $X \subseteq U$ in the neighborhood decision system $<U, C \cup D, N>$, the lower and upper sets of X are:

$$\begin{aligned} \underline{N^k}X &= \{x_i | I(\delta(x_i), X) \geq k, x_i \in U\}, \\ \overline{N^k}X &= \{x_i | I(\delta(x_i), X) \geq 1 - k, x_i \in U\}. \end{aligned} \tag{12}$$

where $1 \geq k \geq 0.5$.

Given these definitions, an easy way to visualize the reasoning for the neighborhood rough set logic is through observing a simple situation as shown in Figure 2. Assume two decision attributes (performance outcomes) d_1 labeled with a "*" and d_2 is labeled with a "+". Taking objects, which in our model represent suppliers, x_1, x_2, and x_3 as examples we assign spherical neighborhoods of a radius $= \delta$ to these objects. We see that $\delta(x_1) \subseteq d_1$, all the other objects within the neighborhood of object x_1 also have the same decision attribute d_1, all represented by "*"; $\delta(x_3) \subseteq d_2$, where all the objects within the neighborhood of object x_3 also have the same decision attribute d_2, all represented by "+". While $\delta(x_2) \cap d_1 \neq \emptyset$ and $\delta(x_2) \cap d_2 \neq \emptyset$, for object x_2, the neighborhood includes objects that have both d_1 and d_2 decision attributes, as exemplified by having both "*" and "+" within its neighborhood. Thus, according to the above definitions, $x_1 \in \underline{N}d_1$, $x_3 \in \underline{N}d_2$, and $x_2 \in BN(D)$. As a whole, regions A1 and A3 are clear lower rough set regions of d_1 and d_2 (i.e. POS(D)), respectively, while A2 is the boundary region of decisions.

Therefore, the neighborhood model divides the samples into two groups: a positive region and a boundary.

Evaluating ecological sustainable performance measures

To help focus the effort on managing the supply chain's economic and ecological performance measures, we introduce a based on the sourcing function of the supply chain operations reference SCOR model (Chae, 2009).

Given the definitions in the previous section on neighborhood rough set characteristics and expressions, we now introduce a procedure within the context of an illustrative application. We will focus on the green and regular supply chain activities associated with the SCOR "sourcing" function which is suppliers performance evaluation.

Example measures and metrics of the "raw" data are shown in Table VII. Historical extreme or ideal ranges from best to worst are also summarized at the bottom of Table VII. We assume an "external" overall business and environmental evaluation scores for each supplier (object) as the decision attributes $D1$ and $D2$, respectively. These outcome

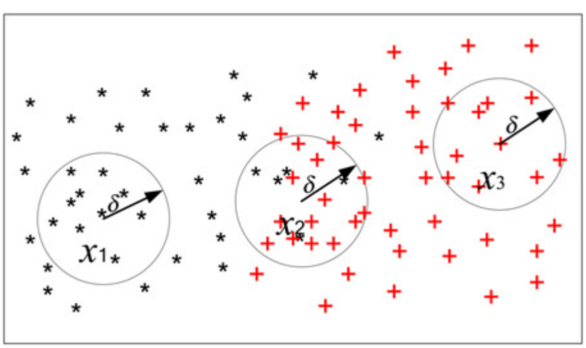

Figure 2.
A visual representation of neighborhood rough set elements and considerations

Table VII.
Evaluation of suppliers on performance measurement by decision makers

Supplier	Cost1	Cost2	CostE	Time1	Time2	TimeE	Qual1	Qual2	QualE	Flex1	Flex2	FlexE	Innv1	Innv2	InnvE	*D*1	*D*2
1	10	[1,5]	23	[23,25]	[11,14]	[4,8]	[0.02,0.09]	VH	VH	VH	212	176	H	2	M	2	2
2	10	[2,8]	7	[−13, − 8]	[42,49]	[23,24]	[0.22,0.73]	H	VH	VH	243	198	VH	2	VH	2	2
3	10	[11,15]	19	[19,21]	[13,16]	[20,23]	[0.5,1.21]	VH	VH	VH	178	186	VH	1	H	2	2
4	9	[9,17]	13	[19,23]	[15,20]	[16,19]	[6.3,7.23]	L	M	VH	321	152	VH	3	H	2	2
5	3	[3,11]	11	[17,20]	[41,45]	[10,12]	[0.89,1.22]	VH	M	H	456	103	M	3	M	2	1
6	8	[37,44]	12	[−16, −12]	[14,20]	[11,13]	[1.23,1.87]	VH	M	VH	73	98	H	4	L	2	1
7	5	[13,17]	10	[16,21]	[12,18]	[10,11]	[1.96,2.56]	H	H	H	249	78	VL	1	M	2	1
8	8	[16,19]	10	[13,17]	[9,15]	[16,19]	[6.32,6.87]	M	L	VH	134	62	H	1	L	2	1
**	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
28	9	[38,45]	20	[19,21]	[18,27]	[1,3]	[7.99,8.72]	VL	VL	VL	46	35	M	4	VL	1	1
29	2	[40,49]	4	[−23, −11]	[55,59]	[21,22]	[8.14,9.78]	VH	VL	VL	469	126	VL	1	L	1	1
30	1	[42,47]	1	[−25, −20]	[42,45]	[2,5]	[1.5,1.9]	VL	VL	VH	423	162	VL	4	VL	1	1

Note: ** Indicates the supplier 09-supplier 27 valuations appear as the last two columns in Table VII. A value of 1 represents a poorer evaluation, while a value of 2 is a better evaluation.

The objects in this neighborhood decision system are represented by 30 suppliers and 17 conditional and decision attributes (performance measures and outcomes).

For each $a_i \in C - Atr$, we use expressions to calculate the dependency degree for each outcome evaluation (D) on a core set of performance measures (Atr) with an additional performance measure when integrated into the Atr set. This dependency degree will be used in the last step to determine the information significance for each added performance measure. Information significance is the amount of unique information or influence a performance measure provides to an evaluation result (D).

After a number of iterations the final set Atr for $\delta = 0.4$, $k = 0.75$ is $\{\text{Flex1}, \text{Time1}, \text{Qual1}, \text{Cost1}\}$ or $\{10, 4, 7, 1\}$. The analysis of these results with for the assigned parameters show that only four core performance measures are needed to fully monitor and evaluate the overall supplier business outcome for the full set of suppliers. The original set of 15 can be reduced to only four remaining performance measures. Thus, the overall outcomes can be effectively monitored for the sourcing function and supplier saving time and effort for both managers and suppliers, who typically have to collect and manage this information. Thus with fewer measures, savings occur and a more direct focus with concentrated efforts to improve these measures make it easier and less costly for the suppliers and management.

Conclusion

In this paper we have provided an overview of how we can evaluate three different green supply chain management practices using rough set theory. In each of these steps we have introduced a different rough set approach. Basic rough set methodology was used to evaluate the supplier selection decision. Rough set theory allows for distillation of a larger set of suppliers into a smaller set of candidate preferred suppliers, and eventually the selection of preferred supplier. Green supplier development programs used a rough set methodology with incomplete information. A neighborhood rough set approach was introduced for joint ecological and business performance measurement management.

Generalizing rough set to incorporate additional measures is not too difficult and management/researchers can easily determine the impacts of these additional measures through case and empirical studies. Overall, the rough set methodology provides some useful insight and support for suppliers management for sustainable supply chains. Ample opportunity exists for building a significant research stream utilizing rough set approaches. For example, the decision making characteristics of rough set can also be applied where weighting schemes can be determined. The linkage of multiple decision makers using various group decision and multiple criteria decision techniques such as the analytical hierarchy process and multiattribute utility theory, may be applicable. This paper provides a comprehensive overview of the various supply chain management environments and helps open many opportunities for this further investigation.

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